PMSCS 623P **Quiz-1** **Summer 2024**

Class Roll: **Name:**

1. What does it mean when we say that an algorithm X is asymptotically more efficient than Y? 1

a) X will be a better choice for all inputs

b) X will be a better choice for all inputs except possibly small inputs

c) X will be a better choice for all inputs except possibly large inputs

d) Y will be a better choice for small inputs

2. In a competition, four different functions are observed. All the functions use a single *for* loop and within the *for* loop, same set of statements are executed. Consider the following for loops: 2

A) for(i = 0; i < n; i++) B) for(i = 0; i < n; i += 2)

C) for(i = 1; i < n; i *= 2) D) for(i = n; i <= n; i /= 2)

If **n** is the size of input (positive), which function is most efficient (if the task to be performed is not an issue)?

3. Analyze the running time of the following code segments and write it using ***O***() notation. 2+2

for(int i=1; i <= n; i=i×3){
  for(int j=n; j<=n; j=j++){
    printf("PMSCS 623");
  }
}
for(k=n/2; k<n/3; k=k++){
      sum++;
}
	int fun(int n)
	{
	  int count = 0;
	  for (int i = 0; i < n; i++)
	    for (int j = i; j > 0; j--)
	      count = count + 1;
	  return count;
	}

4. What is the running time of an insertion sort algorithm if the input array is 1

A = {-3, -3, -3, -3, -3, -3, -3, -3}? **Answer:**

5. Rearrange the following functions (f1, f2, f3, and f4) in increasing order of asymptotic complexity: 2

$f1(n) = n^{(3/2)}$, $f2(n) = 2^n$, $f3(n) = n^{\log(n)}$, $f4(n) = n*\log(n)$

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4. What is the running time of an insertion sort algorithm if the input array is 1

A = {2, 2, 2, 2, 2, 2, 2, 2}? **Answer:**

5. Rearrange the following functions (f1, f2, f3, and f4) in increasing order of asymptotic complexity: 2

$f1(n) = 2^n$, $f2(n) = n^{(3/2)}$, $f3(n) = n*\log(n)$, $f4(n) = n^{\log(n)}$

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